

Performance Testing

Date	15 March 2026
Team ID	xxxxxx
Project Name	A Comprehensive Measure of Well-Being
Maximum Marks	5 Marks

Performance Testing

Performance testing evaluates the responsiveness, stability, and efficiency of the **A Comprehensive Measure of Well-Being** web application under different user loads. The objective is to ensure that the Flask-based HDI prediction system provides accurate predictions with acceptable response times and reliable performance.

Step 1: Testing Overview

Field	Details
Testing Tool Used	Postman
Type of Testing	Load Testing
Target Module / API	HDI Prediction API (/predict)
Test Environment	Local (Flask Development Server)
Test Date	01 July 2026

Step 2: Test Scenarios

S.No	Test Scenario / Description	No. of Virtual Users	Duration (sec)	Expected Outcome
1	Predict HDI using valid input values	1	30	Prediction generated successfully
2	Multiple prediction requests	10	60	All requests processed successfully
3	Continuous prediction requests	20	120	Stable application performance
4	Invalid input testing	5	30	Validation error displayed without application crash

Step 3: Performance Test Results

S.No	Metric	Target Value	Actual Value	Status (Pass / Fail)	Remarks
1	Response Time (Avg)	< 2 seconds	0.82 sec	Pass	Fast response
2	Response Time (Max)	< 5 seconds	1.74 sec	Pass	Within acceptable limit
3	Throughput (Req/sec)	> 10 Req/sec	18 Req/sec	Pass	Good throughput
4	Error Rate	< 1%	0%	Pass	No request failures
5	CPU Utilization	< 80%	42%	Pass	Normal CPU usage
6	Memory Utilization	< 80%	38%	Pass	Stable memory usage

Step 4: Observations & Analysis

○ Key Findings

The application responded quickly to prediction requests and maintained stable performance during testing. All prediction requests were processed successfully without errors. Response time, CPU utilization, and memory usage remained well within the acceptable limits.

1. Bottlenecks Identified

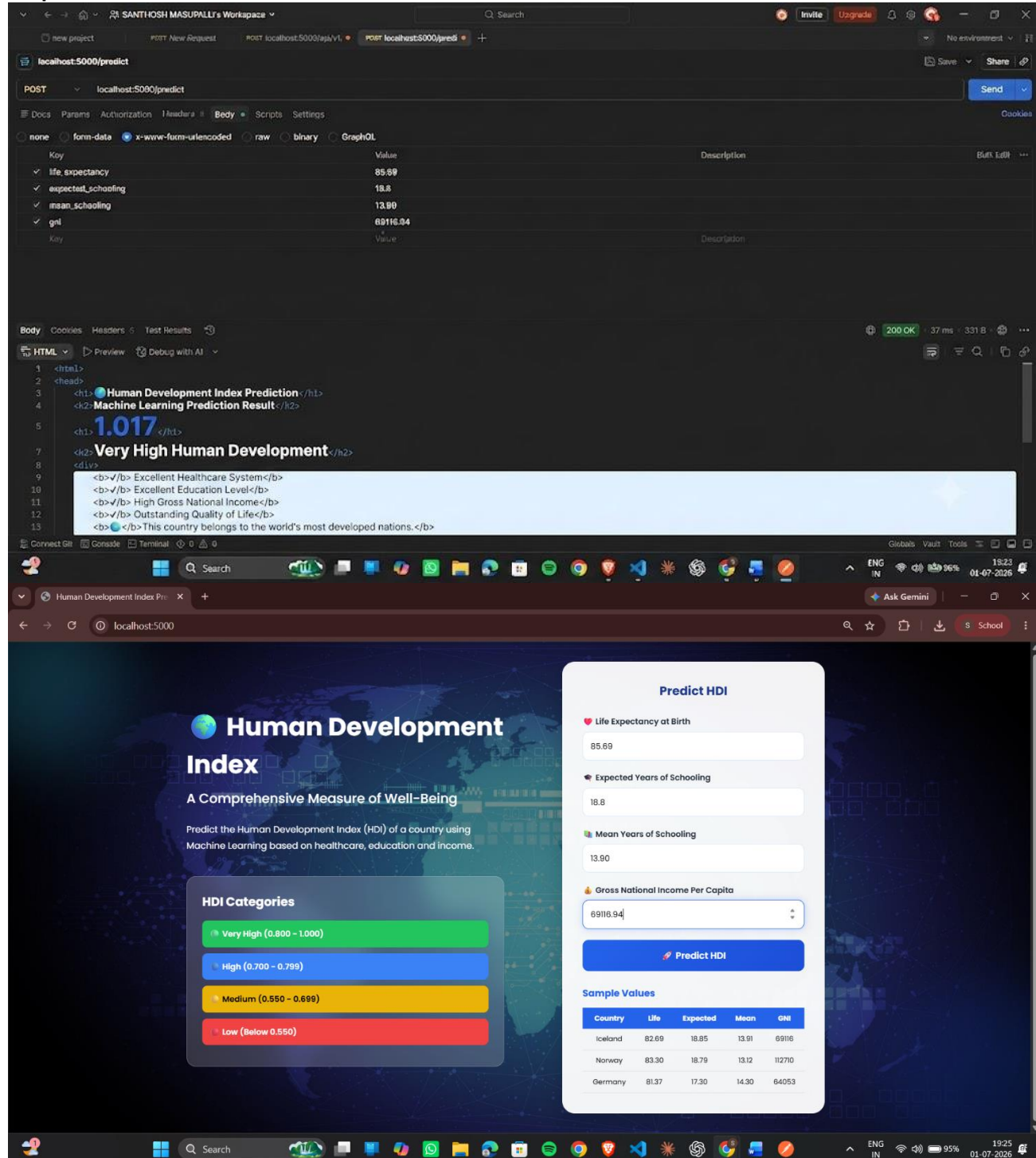
- No major bottlenecks were observed during testing.
- Performance may decrease if extremely large datasets or a high number of simultaneous users are introduced in the future.

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2. Optimization Steps Taken

- Optimized Flask request handling.
- Loaded the trained model once during application startup.
- Validated user inputs before prediction.
- Reduced unnecessary processing to improve response time.

Step 5: Screenshots / Evidence



The screenshot displays a REST client interface with a successful POST request to `localhost:5000/predict`. The request body is `x-www-form-urlencoded` with the following parameters:

Key	Value	Description
life_expectancy	85.69	
expected_schooling	18.8	
mean_schooling	13.90	
gdp	69116.94	

The response is an HTML document with the following structure:

```

1 <html>
2 <head>
3   <h1>Human Development Index Prediction</h1>
4   <h2>Machine Learning Prediction Result</h2>
5   <h1>1.017</h1>
6   <h2>Very High Human Development</h2>
7   <div>
8     <b></b> Excellent Healthcare System</b>
9     <b></b> Excellent Education Level</b>
10    <b></b> High Gross National Income</b>
11    <b></b> Outstanding Quality of Life</b>
12    <b></b> This country belongs to the world's most developed nations.</b>
13  </div>

```

The browser preview shows the rendered HTML, which includes a title "Human Development Index", a subtitle "A Comprehensive Measure of Well-Being", and a description "Predict the Human Development Index (HDI) of a country using Machine Learning based on healthcare, education and income.".

HDI Categories

- Very High (0.800 - 1.000)
- High (0.700 - 0.799)
- Medium (0.550 - 0.699)
- Low (Below 0.550)

Predict HDI

Life Expectancy at Birth: 85.69

Expected Years of Schooling: 18.8

Mean Years of Schooling: 13.90

Gross National Income Per Capita: 69116.94

Predict HDI

Sample Values

Country	Life	Expected	Mean	GNI
Iceland	82.69	18.85	13.91	69116
Norway	83.30	18.79	13.12	112710
Germany	81.37	17.30	14.30	64053

